Total No. of Questions : 4]

SEAT No. :

PB399

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B.E. (Electronics & Telecommunication) (Honors) (Insem)
ARTIFICIAL INTELLIGENCE IN ROBOTICS
(2019 Pattern) (Semester - VIII) (404183 HR)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figure to the right indicates full marks.*
4) *Assume suitable data if necessary.*

Q1) a) What is Artificial Intelligence (AI)? List important AI techniques. **[4]**

b) Explain how to use Artificial Intelligence in manufacturing. **[4]**

c) Explain in detail the Bayesian network method for uncertain reasoning with examples. **[7]**

OR

Q2) a) What are the goals of using AI in the manufacturing sector? **[4]**

b) What are the advantages of using Artificial Intelligence in manufacturing? **[4]**

c) Define Support Vector Machines (SVM), and explain how it works. Enlist its applications. **[7]**

Q3) a) What is Fuzzy Logic? What are the advantages and disadvantages of Fuzzy Logic? **[4]**

b) Explain, Fuzzy Logic in AI with an example. **[4]**

c) What are the types of artificial neural networks (ANN)? Explain the process of ANN training. **[7]**

OR

Q4) a) What is a Genetic Algorithm? Enlist and define basic terminologies used in Genetic algorithms. **[4]**

b) Explain, How does Genetic Algorithm work? **[4]**

c) Explain in detail the K-Means Clustering Algorithm. Write its applications. **[7]**

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